



1. Description

1.1. Project

Project Name	g4-oscilloscope-b
Board Name	NUCLEO-G474RE
Generated with:	STM32CubeMX 6.17.0
Date	06/26/2026

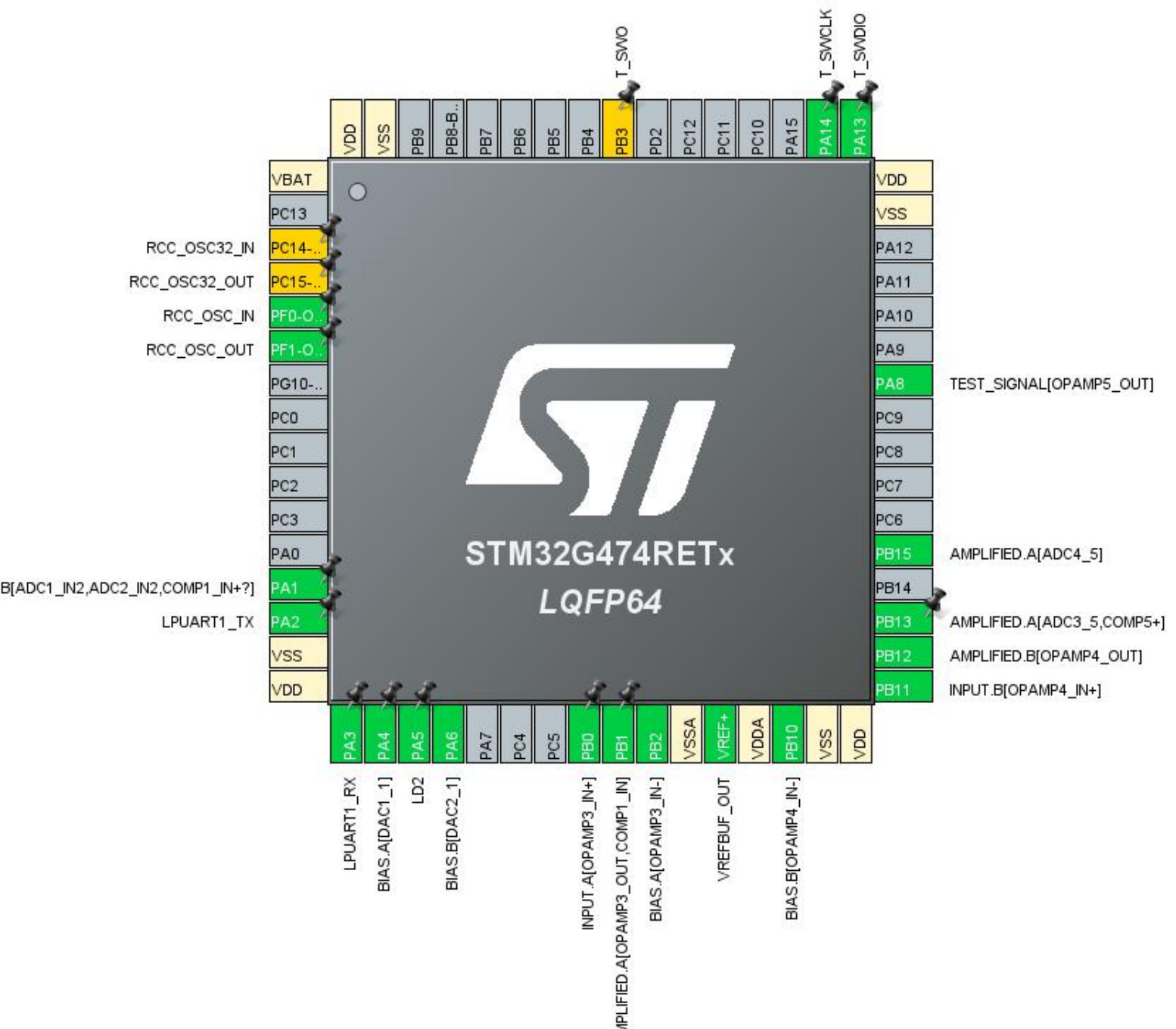
1.2. MCU

MCU Series	STM32G4
MCU Line	STM32G4x4
MCU name	STM32G474RETx
MCU Package	LQFP64
MCU Pin number	64

1.3. Core(s) information

Core(s)	ARM Cortex-M4
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2. Pinout Configuration



3. Pins Configuration

Pin Number LQFP64	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
1	VBAT	Power		
3	PC14-OSC32_IN *	I/O	RCC_OSC32_IN	RCC_OSC32_IN
4	PC15-OSC32_OUT *	I/O	RCC_OSC32_OUT	RCC_OSC32_OUT
5	PF0-OSC_IN	I/O	RCC_OSC_IN	RCC_OSC_IN
6	PF1-OSC_OUT	I/O	RCC_OSC_OUT	RCC_OSC_OUT
13	PA1	I/O	ADC1_IN2, ADC2_IN2	AMPLIFIED.B[ADC1_IN2,A DC2_IN2,COMP1_IN+?]
14	PA2	I/O	LPUART1_TX	
15	VSS	Power		
16	VDD	Power		
17	PA3	I/O	LPUART1_RX	
18	PA4	I/O	DAC1_OUT1	BIAS.A[DAC1_1]
19	PA5 **	I/O	GPIO_Output	LD2
20	PA6	I/O	DAC2_OUT1	BIAS.B[DAC2_1]
24	PB0	I/O	OPAMP3_VINP	INPUT.A[OPAMP3_IN+]
25	PB1	I/O	OPAMP3_VOUT, COMP1_INP	AMPLIFIED.A[OPAMP3_OU T,COMP1_IN]
26	PB2	I/O	OPAMP3_VINM0	BIAS.A[OPAMP3_IN-]
27	VSSA	Power		
28	VREF+	MonolIO	VREFBUF_OUT	
29	VDDA	Power		
30	PB10	I/O	OPAMP4_VINM0	BIAS.B[OPAMP4_IN-]
31	VSS	Power		
32	VDD	Power		
33	PB11	I/O	OPAMP4_VINP	INPUT.B[OPAMP4_IN+]
34	PB12	I/O	OPAMP4_VOUT	AMPLIFIED.B[OPAMP4_OU T]
35	PB13	I/O	COMP5_INP, ADC3_IN5	AMPLIFIED.A[ADC3_5,CO MP5+]
37	PB15	I/O	ADC4_IN5	AMPLIFIED.A[ADC4_5]
42	PA8	I/O	OPAMP5_VOUT	TEST_SIGNAL[OPAMP5_O UT]
47	VSS	Power		
48	VDD	Power		
49	PA13	I/O	SYS_JTMS-SWDIO	T_SWDIO
50	PA14	I/O	SYS_JTCK-SWCLK	T_SWCLK
56	PB3 *	I/O	SYS_JTDO-SWO	T_SWO

Pin Number LQFP64	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
63	VSS	Power		
64	VDD	Power		

** The pin is affected with an I/O function

* The pin is affected with a peripheral function but no peripheral mode is activated

The diagram shows a complex clock tree for the STM32MP157C. It starts with input frequencies: 32.768 kHz (1-1000 kHz), 24 MHz (4-48 MHz), and 12.288 MHz. These inputs feed into various clock sources and dividers. Key components include:

- Input Frequency:** 32.768 kHz (1-1000 kHz) and 24 MHz (4-48 MHz).
- Dividers and Multipliers:** /32, /100, /8, /2, /R, /Q, /5, /P, X1, X170, /1, /2, /4, /8, /16, /32, /64, /128, /256, /512, /1024, /2048, /4096, /8192, /16384, /32768, /65536, /131072, /262144, /524288, /1048576, /2097152, /4194304, /8388608, /16777216, /33554432, /67108864, /134217728, /268435456, /536870912, /1073741824, /2147483648, /4294967296, /8589934592, /17179869184, /34359738368, /68719476736, /137438953472, /274877906944, /549755813888, /1099511627776, /2199023255552, /4398046511104, /8796093022208, /17592186044416, /35184372088832, /70368744177664, /140737488355328, /281474976710656, /562949953421312, /1125899906842624, /2251799813685248, /4503599627370496, /9007199254740992, /18014398509481984, /36028797018963968, /72057594037927936, /144115188075855872, /288230376151711744, /576460752303423488, /1152921504606846976, /2305843009213693952, /4611686018427387904, /9223372036854775808, /18446744073709551616, /36893488147419103232, /73786976294838206464, /147573952589676412928, /295147905179352825856, /590295810358705651712, /1180591620717411303424, /2361183241434822606848, /4722366482869645213696, /9444732965739290427392, /18889465931478580854784, /37778931862957161709568, /75557863725914323419136, /151115727451828646838272, /302231454903657293676544, /604462909807314587353088, /1208925819614629174706176, /2417851639229258349412352, /4835703278458516698824704, /9671406556917033397649408, /19342813113834066795298816, /38685626227668133590597632, /77371252455336267181195264, /154742504910672534362390528, /309485009821345068724781056, /618970019642690137449562112, /1237940039285380274899124224, /2475880078570760549798248448, /4951760157141521099596496896, /9903520314283042199192993792, /19807040628566084398385987584, /39614081257132168796771975168, /79228162514264337593543950336, /158456325028528675187087900672, /316912650057057350374175801344, /633825300114114700748351602688, /1267650600228229401496703205376, /2535301200456458802993406410752, /5070602400912917605986812821504, /10141204801825835211973625643008, /20282409603651670423947251286016, /40564819207303340847894502572032, /81129638414606681695789005144064, /162259276829213363391578010288128, /324518553658426726783156020576256, /649037107316853453566312041152512, /1298074214633706907132624082305024, /2596148429267413814265248164610048, /5192296858534827628530496329220096, /10384593717069655257060992658440192, /20769187434139310514121985316880384, /41538374868278621028243970633760768, /83076749736557242056487941267521536, /166153499473114484112975882535043072, /332306998946228968225951765070086144, /664613997892457936451903530140172288, /1329227995784915872903807060280344576, /2658455991569831745807614120560689152, /5316911983139663491615228241121378304, /10633823966279326983230456482242756608, /21267647932558653966460912964485513216, /42535295865117307932921825928971026432, /85070591730234615865843651857942052864, /170141183460469231731687303715884105728, /340282366920938463463374607431768211456, /680564733841876926926749214863536422912, /1361129467683753853853498429727072845824, /2722258935367507707706996859454145691648, /5444517870735015415413993718908291383296, /10889035741470030830827987437816582766592, /21778071482940061661655974875633165533184, /43556142965880123323311949751266331066368, /87112285931760246646623899502532662132736, /174224571863520493293247799005065324265472, /348449143727040986586495598010130648530944, /696898287454081973172991196020261297061888, /1393796574908163946345982392040522594123776, /2787593149816327892691964784081045188247552, /5575186299632655785383929568162090376495104, /11150372599265311570767859136324180752990208, /22300745198530623141535718272648361505980416, /44601490397061246283071436545296723011960832, /89202980794122492566142873090593446023921664, /178405961588244985132285746181186892047843328, /356811923176489970264571492362373784095686656, /713623846352979940529142984724747568191373312, /1427247692705959881058285969449495136382746624, /2854495385411919762116

1. Power Consumption Calculator report

1.1. Microcontroller Selection

Series	STM32G4
Line	STM32G4x4
MCU	STM32G474RETx
Datasheet	DS12288 Rev0

1.2. Parameter Selection

Temperature	25
Vdd	3.0

1.3. Battery Selection

Battery	Li-SOCL2(A3400)
Capacity	3400.0 mAh
Self Discharge	0.08 %/month
Nominal Voltage	3.6 V
Max Cont Current	100.0 mA
Max Pulse Current	200.0 mA
Cells in series	1
Cells in parallel	1

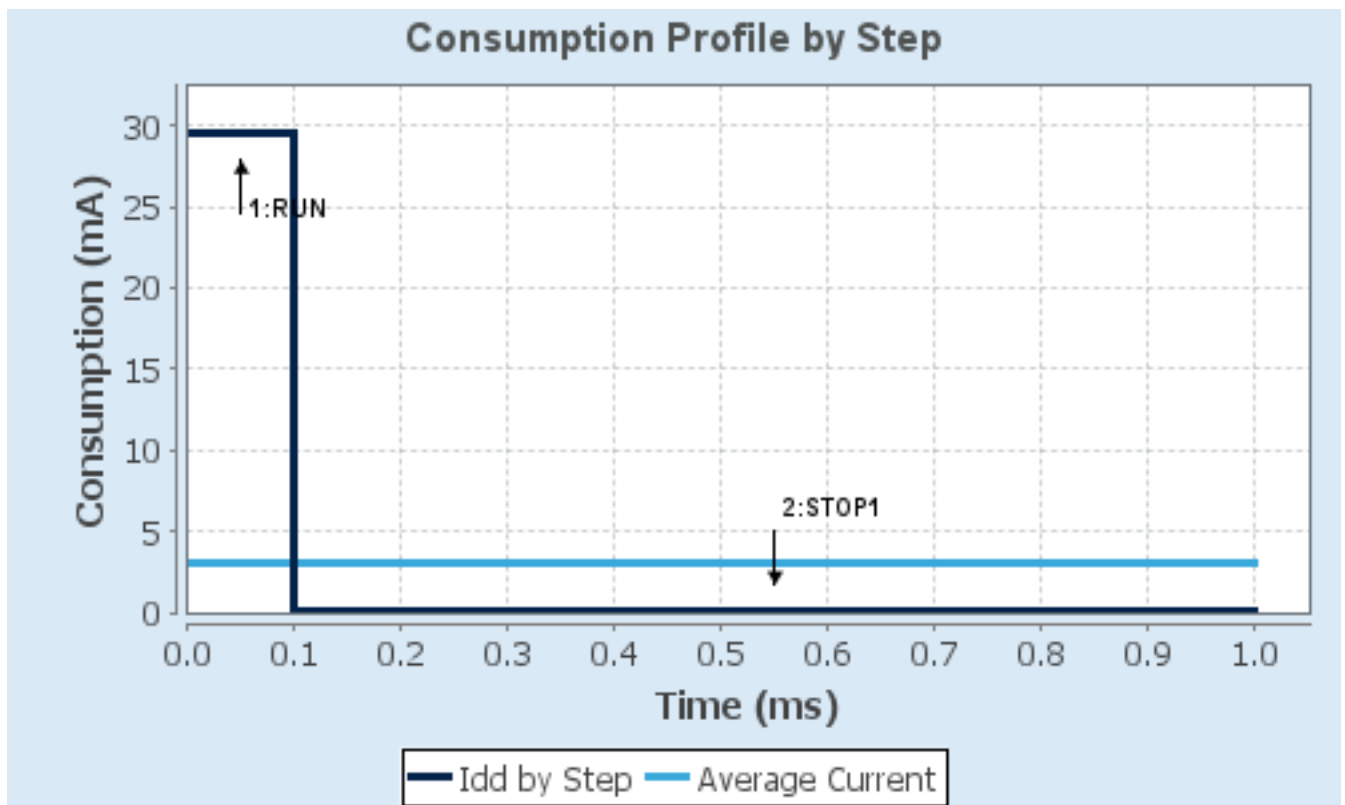
1.4. Sequence

Step	Step1	Step2
Mode	RUN	STOP1
Vdd	3.0	3.0
Voltage Source	Battery	Battery
Range	Range1-Boost	NoRange
Fetch Type	FLASH/DualBank/ART	NA
CPU Frequency	170 MHz	0 Hz
Clock Configuration	HSE BYP PLL	ALL CLOCKS OFF
Clock Source Frequency	4 MHz	0 Hz
Peripherals		
Additional Cons.	0 mA	0 mA
Average Current	29.5 mA	80.5 μ A
Duration	0.1 ms	0.9 ms
DMIPS	213.0	0.0
Ta Max	124.25	129.98
Category	In DS Table	In DS Table

1.5. Results

Sequence Time	1 ms	Average Current	3.02 mA
Battery Life	1 month, 16 days, 9 hours	Average DMIPS	212.5 DMIPS

1.6. Chart



2. Software Project

2.1. Project Settings

Name	Value
Project Name	g4-oscilloscope-b
Project Folder	D:\projects\g4-oscilloscope-b
Toolchain / IDE	CMake
Firmware Package Name and Version	STM32Cube FW_G4 V1.6.3
Application Structure	Advanced
Generate Under Root	No
Do not generate the main()	No
Minimum Heap Size	0x200
Minimum Stack Size	0x400

2.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy only the necessary library files
Generate peripheral initialization as a pair of '.c/.h' files	Yes
Backup previously generated files when re-generating	No
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	Yes
Set all free pins as analog (to optimize the power consumption)	No
Enable Full Assert	No

2.3. Advanced Settings - Generated Function Calls

Rank	Function Name	Peripheral Instance Name
1	MX_GPIO_Init	GPIO
2	MX_DMA_Init	DMA
3	SystemClock_Config	RCC
4	MX_ADC3_Init	ADC3
5	MX_OPAMP3_Init	OPAMP3
6	MX_DAC1_Init	DAC1
7	MX_ADC4_Init	ADC4
8	MX_LPUART1_UART_Init	LPUART1
9	MX_TIM1_Init	TIM1
10	MX_TIM15_Init	TIM15
11	MX_TIM2_Init	TIM2

Rank	Function Name	Peripheral Instance Name
12	MX_COMP1_Init	COMP1
13	MX_DAC4_Init	DAC4
14	MX_OPAMP5_Init	OPAMP5
15	MX_DAC3_Init	DAC3
16	MX_ADC1_Init	ADC1
17	MX_ADC2_Init	ADC2
18	MX_DAC2_Init	DAC2
19	MX_OPAMP4_Init	OPAMP4

3. Peripherals and Middlewares Configuration

3.1. ADC1

IN2: IN2 Single-ended

3.1.1. Parameter Settings:

ADCs_Common_Settings:

Mode

Dual interleaved mode only *

DMA Access Mode

DMA access mode enabled

Delay between 2 sampling phases

6 Cycles *

ADC_Settings:

Clock Prescaler

Asynchronous clock mode divided by 1 *

Resolution

ADC 12-bit resolution

Data Alignment

Right alignment

Gain Compensation

0

Scan Conversion Mode

Disabled

End Of Conversion Selection

End of single conversion

Low Power Auto Wait

Disabled

Continuous Conversion Mode

Disabled

Discontinuous Conversion Mode

Disabled

DMA Continuous Requests

Enabled *

Overrun behaviour

Overrun data overwritten *

ADC_Regular_ConversionMode:

Enable Regular Conversions

Enable

Enable Regular Oversampling

Disable

Number Of Conversion

1

External Trigger Conversion Source

Timer 2 Trigger Out event *

External Trigger Conversion Edge

Trigger detection on the rising edge

Rank

1

Channel

Channel 2

Sampling Time

2.5 Cycles

Offset Number

No offset

ADC_Injected_ConversionMode:

Enable Injected Conversions

Disable

Analog Watchdog 1:

Enable Analog WatchDog1 Mode

false

Analog Watchdog 2:

Enable Analog WatchDog2 Mode

false

Analog Watchdog 3:

Enable Analog WatchDog3 Mode

false

3.2. ADC2

IN2: IN2 Single-ended

3.2.1. Parameter Settings:

ADCs_Common_Settings:

Mode

Dual interleaved mode only *

DMA Access Mode

DMA access mode enabled

Delay between 2 sampling phases

6 Cycles *

ADC_Settings:

Clock Prescaler

Asynchronous clock mode divided by 1 *

Resolution

ADC 12-bit resolution

Data Alignment

Right alignment

Gain Compensation

0

Scan Conversion Mode

Disabled

End Of Conversion Selection

End of single conversion

Low Power Auto Wait

Disabled

Continuous Conversion Mode

Disabled

Discontinuous Conversion Mode

Disabled

DMA Continuous Requests

Disabled

Overrun behaviour

Overrun data overwritten *

ADC_Regular_ConversionMode:

Enable Regular Conversions

Enable

Enable Regular Oversampling

Disable

Number Of Conversion

1

Rank

1

Channel

Channel 2

Sampling Time

2.5 Cycles

Offset Number

No offset

ADC_Injected_ConversionMode:

Enable Injected Conversions

Disable

Analog Watchdog 1:

Enable Analog WatchDog1 Mode

false

Analog Watchdog 2:

Enable Analog WatchDog2 Mode

false

Analog Watchdog 3:

Enable Analog WatchDog3 Mode

false

3.3. ADC3

mode: IN5

3.3.1. Parameter Settings:

ADCs_Common_Settings:

Mode

Dual interleaved mode only *

DMA Access Mode

DMA access mode enabled

Delay between 2 sampling phases

6 Cycles *

ADC_Settings:

Clock Prescaler

Asynchronous clock mode divided by 1 *

Resolution

ADC 12-bit resolution

Data Alignment

Right alignment

Gain Compensation

0

Scan Conversion Mode

Disabled

End Of Conversion Selection

End of single conversion

Low Power Auto Wait

Disabled

Continuous Conversion Mode

Disabled

Discontinuous Conversion Mode

Disabled

DMA Continuous Requests

Enabled *

Overrun behaviour

Overrun data overwritten *

ADC_Regular_ConversionMode:

Enable Regular Conversions

Enable

Enable Regular Oversampling

Disable

Number Of Conversion

1

External Trigger Conversion Source

Timer 2 Trigger Out event *

External Trigger Conversion Edge

Trigger detection on the rising edge

Rank

1

Channel

Channel 5

Sampling Time

2.5 Cycles

Offset Number

No offset

ADC_Injected_ConversionMode:

Enable Injected Conversions

Disable

Analog Watchdog 1:

Enable Analog WatchDog1 Mode

false

Analog Watchdog 2:

Enable Analog WatchDog2 Mode

false

Analog Watchdog 3:

Enable Analog WatchDog3 Mode false

3.4. ADC4

mode: IN5

3.4.1. Parameter Settings:

ADCs_Common_Settings:

Mode

DMA Access Mode

Delay between 2 sampling phases

ADC_Settings:

Clock Prescaler

Resolution

Data Alignment

Gain Compensation

Scan Conversion Mode

End Of Conversion Selection

Low Power Auto Wait

Continuous Conversion Mode

Discontinuous Conversion Mode

DMA Continuous Requests

Overrun behaviour

ADC_Regular_ConversionMode:

Enable Regular Conversions

Enable Regular Oversampling

Number Of Conversion

Rank

Channel

Sampling Time

Offset Number

ADC_Injected_ConversionMode:

Enable Injected Conversions

Analog Watchdog 1:

Enable Analog WatchDog1 Mode

Analog Watchdog 2:

Enable Analog WatchDog2 Mode

Analog Watchdog 3:

Enable Analog WatchDog3 Mode

Dual interleaved mode only *

DMA access mode enabled

6 Cycles *

Asynchronous clock mode divided by 1 *

ADC 12-bit resolution

Right alignment

0

Disabled

End of single conversion

Disabled

Disabled

Disabled

Disabled

Overrun data overwritten *

Enable

Disable

1

1

Channel 5

2.5 Cycles

No offset

Disable

false

false

false

3.5. COMP1

mode: Input [+]

Input [-]: DAC3 OUT1

3.5.1. Parameter Settings:

Basic Parameters:

Trigger Mode

Rising/Falling Edge Interrupt *

Hysteresis Level

Level 10mV *

Output Configuration:

Blanking Source

None

Output Polarity

COMP output on GPIO isn't inverted

3.6. DAC1

OUT1 mode: Connected to external pin only

3.6.1. Parameter Settings:

DAC Out1 Settings:

Mode selected

Normal Mode

Output Buffer

Enable

DAC High Frequency

Mode Automatic

DMA Double Data

Disable

Signed Format

Disable

Trigger

None

Trigger2

None

User Trimming

Factory trimming

3.7. DAC2

OUT1 mode: Connected to external pin only

3.7.1. Parameter Settings:

DAC Out1 Settings:

Mode selected

Normal Mode

Output Buffer

Enable

DAC High Frequency

Mode Automatic

DMA Double Data	Disable
Signed Format	Disable
Trigger	None
Trigger2	None
User Trimming	Factory trimming

3.8. DAC3

mode: OUT1 mode

3.8.1. Parameter Settings:

DAC Out1 Settings:

Mode selected	Normal Mode
Output Buffer	Disable
DAC High Frequency	Mode Automatic
DMA Double Data	Disable
Signed Format	Disable
Trigger	None
Trigger2	None
User Trimming	Factory trimming

3.9. DAC4

mode: OUT2 mode

3.9.1. Parameter Settings:

DAC Out2 Settings:

Mode selected	Normal Mode
Output Buffer	Disable
DAC High Frequency	Mode Automatic
DMA Double Data	Disable
Signed Format	Disable
Trigger	Timer 15 Trigger Out event *
Trigger2	None
Wave generation mode	Disabled
User Trimming	Factory trimming

3.10. LPUART1

Mode: Asynchronous

3.10.1. Parameter Settings:

Basic Parameters:

Baud Rate	460800 *
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	FIFO mode enable *
Txfifo Threshold	half full configuration *
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.11. OPAMP3

Mode: PGA Connected-INVERTINGINPUT_IO0_BIAS

3.11.1. Parameter Settings:

Basic Parameters:

Power Mode	High Speed *
PGA Gain	16 or -15 *
User Trimming	Disable

3.12. OPAMP4

Mode: PGA Connected-INVERTINGINPUT_IO0_BIAS

3.12.1. Parameter Settings:

Basic Parameters:

Power Mode	Normal
PGA Gain	2 or -1
User Trimming	Disable

3.13. OPAMP5

Mode: Follower-DAC4_OUT2-INP

3.13.1. Parameter Settings:

Basic Parameters:

Power Mode	Normal
User Trimming	Disable

3.14. RCC

High Speed Clock (HSE): Crystal/Ceramic Resonator

3.14.1. Parameter Settings:

System Parameters:

VDD voltage (V)	3.3
Instruction Cache	Enabled
Prefetch Buffer	Disabled
Data Cache	Enabled
Flash Latency(WS)	4 WS (5 CPU cycle)

RCC Parameters:

HSI Calibration Value	64
HSE Startup Timeout Value (ms)	100
LSE Startup Timeout Value (ms)	5000

Power Parameters:

Power Regulator Voltage Scale	Power Regulator Voltage Scale 1
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Peripherals Clock Configuration:

Generate the peripherals clock configuration	TRUE
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3.15. SYS

Debug: Serial Wire

VREFBUF Mode: Internal voltage reference

Timebase Source: TIM7

mode: save power of non-active UCPD - deactive Dead Battery pull-up

3.15.1. Parameter Settings:

Voltage_Reference_Buffer_Settings:

Trimming Mode	Factory Trimming
Internal Voltage reference scale	SCALE 0: around 2.048 V

3.16. TIM1

Slave Mode: External Clock Mode 1

Trigger Source: ITR1

Channel1: PWM Generation No Output

mode: One Pulse Mode

3.16.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Dithering	Disable
Counter Period (AutoReload Register - 16 bits value)	65535
Internal Clock Division (CKD)	No Division
Repetition Counter (RCR - 16 bits value)	0
auto-reload preload	Disable
Slave Mode Controller	ETR mode 1

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Output Compare (OC1REF) *
Trigger Event Selection TRGO2	Reset (UG bit from TIMx_EGR)

Break And Dead Time management - BRK Configuration:

BRK State	Disable
BRK Polarity	High
BRK Filter (4 bits value)	0
BRK Sources Configuration	
- Digital Input	Disable

- COMP1	Disable
- COMP2	Disable
- COMP3	Disable
- COMP4	Disable
- COMP5	Disable
- COMP6	Disable
- COMP7	Disable

Break And Dead Time management - BRK2 Configuration:

BRK2 State	Disable
BRK2 Polarity	High
BRK2 Filter (4 bits value)	0
BRK2 Sources Configuration	
- Digital Input	Disable
- COMP1	Disable
- COMP2	Disable
- COMP3	Disable
- COMP4	Disable
- COMP5	Disable
- COMP6	Disable
- COMP7	Disable

Break And Dead Time management - Output Configuration:

Automatic Output State	Disable
Off State Selection for Run Mode (OSSR)	Disable
Off State Selection for Idle Mode (OSSI)	Disable
Lock Configuration	Off

Clear Input:

Clear Input Source	Disable
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PWM Generation Channel 1:

Mode	PWM mode 1
Pulse (16 bits value)	99 *
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CH Idle State	Set *

3.17. TIM2

Slave Mode: Gated Mode

Trigger Source: ITR0

Clock Source : Internal Clock

3.17.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	39 *
Counter Mode	Up
Dithering	Disable
Counter Period (AutoReload Register - 32 bits value)	19 *
Internal Clock Division (CKD)	No Division
auto-reload preload	Disable
Slave Mode Controller	Gated Mode

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Update Event *

3.18. TIM15

mode: Clock Source

3.18.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	30 *
Counter Mode	Up
Dithering	Disable
Counter Period (AutoReload Register - 16 bits value)	16 *
Internal Clock Division (CKD)	No Division
Repetition Counter (RCR - 8 bits value)	0
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection	Update Event *

3.19. FREERTOS

Interface: CMSIS_V2

3.19.1. Config parameters:

API:

FreeRTOS API CMSIS v2

Versions:

FreeRTOS version 10.3.1
CMSIS-RTOS version 2.1.3

MPU/FPU:

ENABLE_MPU Disabled
ENABLE_FPU Disabled

Kernel settings:

USE_PREEMPTION Enabled
CPU_CLOCK_HZ SystemCoreClock
TICK_RATE_HZ 1000
MAX_PRIORITIES 56
USE_SB_COMPLETED_CALLBACK 0
USE_MINI_LIST_ITEM 1
MINIMAL_STACK_SIZE 128
MAX_TASK_NAME_LEN 16
USE_16_BIT_TICKS Disabled
IDLE_SHOULD_YIELD Enabled
USE_MUTEXES Enabled
USE_RECURSIVE_MUTEXES Enabled
USE_COUNTING_SEMAPHORES Enabled
QUEUE_REGISTRY_SIZE 8
USE_APPLICATION_TASK_TAG Disabled
ENABLE_BACKWARD_COMPATIBILITY Enabled
USE_PORT_OPTIMISED_TASK_SELECTION Disabled
USE_TICKLESS_IDLE Disabled
USE_TASK_NOTIFICATIONS Enabled
RECORD_STACK_HIGH_ADDRESS Disabled

Memory management settings:

Memory Allocation Dynamic / Static
TOTAL_HEAP_SIZE **30720 ***
Memory Management scheme heap_4

Hook function related definitions:

USE_IDLE_HOOK Disabled
USE_TICK_HOOK Disabled
USE_MALLOC_FAILED_HOOK Disabled
USE_DAEMON_TASK_STARTUP_HOOK Disabled
CHECK_FOR_STACK_OVERFLOW Disabled

Run time and task stats gathering related definitions:

GENERATE_RUN_TIME_STATS **Enabled ***
USE_TRACE_FACILITY Enabled

USE_STATS_FORMATTING_FUNCTIONS Disabled

Co-routine related definitions:

USE_CO_ROUTINES Disabled

MAX_CO_ROUTINE_PRIORITIES 2

Software timer definitions:

USE_TIMERS Enabled

TIMER_TASK_PRIORITY 2

TIMER_QUEUE_LENGTH 10

TIMER_TASK_STACK_DEPTH 256

Interrupt nesting behaviour configuration:

LIBRARY_LOWEST_INTERRUPT_PRIORITY 15

LIBRARY_MAX_SYSCALL_INTERRUPT_PRIORITY 5

Added with 10.2.1 support:

MESSAGE_BUFFER_LENGTH_TYPE size_t

USE_POSIX_ERRNO Disabled

CMSIS-RTOS V2 flags:

USE_OS2_THREAD_SUSPEND_RESUME Enabled

USE_OS2_THREAD_ENUMERATE Enabled

USE_OS2_EVENTFLAGS_FROM_ISR Enabled

USE_OS2_THREAD_FLAGS Enabled

USE_OS2_TIMER Enabled

USE_OS2_MUTEX Enabled

3.19.2. Include parameters:

Include definitions:

vTaskPrioritySet Enabled

uxTaskPriorityGet Enabled

vTaskDelete Enabled

vTaskCleanUpResources Disabled

vTaskSuspend Enabled

vTaskDelayUntil Enabled

vTaskDelay Enabled

xTaskGetSchedulerState Enabled

xTaskResumeFromISR Enabled

xQueueGetMutexHolder Enabled

xSemaphoreGetMutexHolder Disabled

pcTaskGetTaskName Disabled

uxTaskGetStackHighWaterMark Enabled

xTaskGetCurrentTaskHandle Enabled

eTaskGetState Enabled

xEventGroupSetBitFromISR	Disabled
xTimerPendFunctionCall	Enabled
xTaskAbortDelay	Disabled
xTaskGetHandle	Disabled
uxTaskGetStackHighWaterMark2	Disabled

3.19.3. Advanced settings:

Newlib settings (see parameter description first):

USE_NEWLIB_REENTRANT	Disabled
----------------------	----------

Project settings (see parameter description first):

Use FW pack heap file	Enabled
-----------------------	---------

*** User modified value**

4. System Configuration

4.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
ADC1	PA1	ADC1_IN2	Analog mode	No pull-up and no pull-down	n/a	AMPLIFIED.B[ADC1_IN2, ADC2_IN2,COMP1_IN+?]
ADC2	PA1	ADC2_IN2	Analog mode	No pull-up and no pull-down	n/a	AMPLIFIED.B[ADC1_IN2, ADC2_IN2,COMP1_IN+?]
ADC3	PB13	ADC3_IN5	Analog mode	No pull-up and no pull-down	n/a	AMPLIFIED.A[ADC3_5,COMP5+]
ADC4	PB15	ADC4_IN5	Analog mode	No pull-up and no pull-down	n/a	AMPLIFIED.A[ADC4_5]
COMP1	PB1	COMP1_INP	Analog mode	No pull-up and no pull-down	n/a	AMPLIFIED.A[OPAMP3_OUT,COMP1_IN]
DAC1	PA4	DAC1_OUT1	Analog mode	No pull-up and no pull-down	n/a	BIAS.A[DAC1_1]
DAC2	PA6	DAC2_OUT1	Analog mode	No pull-up and no pull-down	n/a	BIAS.B[DAC2_1]
LPUART1	PA2	LPUART1_TX	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
	PA3	LPUART1_RX	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
OPAMP3	PB0	OPAMP3_VINP	Analog mode	No pull-up and no pull-down	n/a	INPUT.A[OPAMP3_IN+]
	PB1	OPAMP3_VOUT	Analog mode	No pull-up and no pull-down	n/a	AMPLIFIED.A[OPAMP3_OUT,COMP1_IN]
	PB2	OPAMP3_VINM0	Analog mode	No pull-up and no pull-down	n/a	BIAS.A[OPAMP3_IN-]
OPAMP4	PB10	OPAMP4_VINM0	Analog mode	No pull-up and no pull-down	n/a	BIAS.B[OPAMP4_IN-]
	PB11	OPAMP4_VINP	Analog mode	No pull-up and no pull-down	n/a	INPUT.B[OPAMP4_IN+]
	PB12	OPAMP4_VOUT	Analog mode	No pull-up and no pull-down	n/a	AMPLIFIED.B[OPAMP4_OUT]
OPAMP5	PA8	OPAMP5_VOUT	Analog mode	No pull-up and no pull-down	n/a	TEST_SIGNAL[OPAMP5_OUT]
RCC	PF0-OSC_IN	RCC_OSC_IN	n/a	n/a	n/a	RCC_OSC_IN
	PF1-OSC_OUT	RCC_OSC_OUT	n/a	n/a	n/a	RCC_OSC_OUT
SYS	VREF+	VREFBUF_OUT	n/a	n/a	n/a	
	PA13	SYS_JTMS-SWDIO	n/a	n/a	n/a	T_SWDIO
	PA14	SYS_JTCK-SWCLK	n/a	n/a	n/a	T_SWCLK
Single Mapped Signals	PC14-OSC32_IN	RCC_OSC32_IN	n/a	n/a	n/a	RCC_OSC32_IN
	PC15-OSC32_OUT	RCC_OSC32_OUT	n/a	n/a	n/a	RCC_OSC32_OUT

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
	PB3	SYS_JTDO- SWO	n/a	n/a	n/a	T_SWO
GPIO	PA5	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	LD2

4.2. DMA configuration

DMA request	Stream	Direction	Priority
MEMTOMEM	DMA1_Channel2	Memory To Memory	Low
ADC3	DMA1_Channel3	Peripheral To Memory	Low
LPUART1_TX	DMA1_Channel4	Memory To Peripheral	Low
DAC4_CH2	DMA1_Channel1	Memory To Peripheral	Low
ADC1	DMA1_Channel5	Peripheral To Memory	Low
MEMTOMEM	DMA1_Channel6	Memory To Memory	Low

MEMTOMEM: DMA1_Channel2 DMA request Settings:

Mode: Normal
 Src Memory Increment: **Enable ***
 Dst Memory Increment: **Enable ***
 Src Memory Data Width: **Word ***
 Dst Memory Data Width: **Word ***

ADC3: DMA1_Channel3 DMA request Settings:

Mode: **Circular ***
 Peripheral Increment: Disable
 Memory Increment: **Enable ***
 Peripheral Data Width: **Word ***
 Memory Data Width: **Word ***

LPUART1_TX: DMA1_Channel4 DMA request Settings:

Mode: Normal
 Peripheral Increment: Disable
 Memory Increment: **Enable ***
 Peripheral Data Width: Byte
 Memory Data Width: Byte

DAC4_CH2: DMA1_Channel1 DMA request Settings:

Mode: **Circular ***
 Peripheral Increment: Disable

Memory Increment: **Enable ***
Peripheral Data Width: Word
Memory Data Width: Half Word

ADC1: DMA1_Channel5 DMA request Settings:

Mode: **Circular ***
Peripheral Increment: Disable
Memory Increment: **Enable ***
Peripheral Data Width: **Word ***
Memory Data Width: **Word ***

MENTOMEM: DMA1_Channel6 DMA request Settings:

Mode: Normal
Src Memory Increment: **Enable ***
Dst Memory Increment: **Enable ***
Src Memory Data Width: **Word ***
Dst Memory Data Width: **Word ***

4.3. NVIC configuration

4.3.1. NVIC

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Prefetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	15	0
System tick timer	true	15	0
DMA1 channel1 global interrupt	true	5	0
DMA1 channel2 global interrupt	true	5	0
DMA1 channel3 global interrupt	true	5	0
DMA1 channel4 global interrupt	true	5	0
DMA1 channel5 global interrupt	true	5	0
TIM1 capture compare interrupt	true	5	0
TIM7 global interrupt, DAC2 and DAC4 channel underrun error interrupts	true	15	0
COMP1, COMP2 and COMP3 interrupts through EXTI lines 21, 22 and 29	true	5	0
LPUART1 global interrupt	true	5	0
PVD/PVM1/PVM2/PVM3/PVM4 interrupts through EXTI lines 16/38/39/40/41	unused		
Flash global interrupt	unused		
RCC global interrupt	unused		
DMA1 channel6 global interrupt	unused		
ADC1 and ADC2 global interrupt	unused		
TIM1 break interrupt and TIM15 global interrupt	unused		
TIM1 update interrupt and TIM16 global interrupt	unused		
TIM1 trigger and commutation interrupts and TIM17 global interrupt	unused		
TIM2 global interrupt	unused		
ADC3 global interrupt	unused		
TIM6 global interrupt, DAC1 and DAC3 channel underrun error interrupts	unused		
ADC4 global interrupt	unused		
FPU global interrupt	unused		

4.3.2. NVIC Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false
Hard fault interrupt	false	true	false
Memory management fault	false	true	false
Prefetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
System service call via SWI instruction	false	false	false
Debug monitor	false	true	false
Pendable request for system service	false	false	false
System tick timer	false	false	true
DMA1 channel1 global interrupt	false	true	true
DMA1 channel2 global interrupt	false	true	true
DMA1 channel3 global interrupt	false	true	true
DMA1 channel4 global interrupt	false	true	true
DMA1 channel5 global interrupt	false	true	true
TIM1 capture compare interrupt	false	true	true
TIM7 global interrupt, DAC2 and DAC4 channel underrun error interrupts	false	true	true
COMP1, COMP2 and COMP3 interrupts through EXTI lines 21, 22 and 29	false	true	true
LPUART1 global interrupt	false	true	true

*** User modified value**

5. System Views

5.1. Category view

5.1.1. Current

Middleware								
FREERTOS								
System Core	Analog		Timers	Connectivity	Multimedia	Security	Computing	Utilities
DMA	ADC1	ADC2	TIM1	LPUART1				
GPIO	ADC3	ADC4	TIM2					
IVVIC	COMP1	DAC1	TIM15					
RCC	DAC2	DAC3						
SYS	DAC4	OPAMP3						
	OPAMP4	OPAMP5						

6. Docs & Resources

Type	Link
BSDL files	https://www.st.com/resource/en/bsdl_model/stm32g4_bsd1.zip
IBIS models	https://www.st.com/resource/en/ibis_model/stm32g4_ibis.zip
System View Description	https://www.st.com/resource/en/svd/stm32g4_svd.zip
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_eval_tools_portfolio.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_stm8_functional-safety-packages.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-usb-c-pd-solutions-presentation.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32g4-series-product-overview.pdf
Brochures	https://www.st.com/resource/en/brochure/products-and-solutions-for-plcs-and-smart-i-os.pdf
Brochures	https://www.st.com/resource/en/brochure/expansion-boards-for-intelligent-power-switches.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32g4.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32nucleo.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32trust.pdf
Flyers	https://www.st.com/resource/en/flyer/flpowerstbd.pdf
Flyers	https://www.st.com/resource/en/flyer/fldpstpf11120.pdf
Security Bulletin	https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-

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Application Notes https://www.st.com/resource/en/application_note/an3156-usb-dfu-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf

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for related Tools obtaining-ulcsaiec-607301603351-class-b-certification-in-any-stm32-
& Software application-stmicroelectronics.pdf

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for related Tools inapplication-programming-iap-using-the-usart-stmicroelectronics.pdf
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for related Tools processing-for-stm32-microcontrollers-using-cmsis-stmicroelectronics.pdf
& Software

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for related Tools guide-for-the-xcubesbsfu-stm32cube-expansion-package-
& Software stmicroelectronics.pdf

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for related Tools implementation-with-the-fmac-using-stm32cubeg4-mcu-package-
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for related Tools firmware-examples-for-stm32g4-series-stmicroelectronics.pdf
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for related Tools rgb-led-control-using-the-bg474edpow1-discovery-kit-
& Software stmicroelectronics.pdf

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for related Tools with-projects-based-on-the-stm32mp1-series-in-stm32cubeide-
& Software stmicroelectronics.pdf

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& Software stm32cubeide-stmicroelectronics.pdf

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for related Tools with-projects-based-on-the-stm32l5-series-in-stm32cubeide-
& Software stmicroelectronics.pdf

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for related Tools simple-usbp-d-sink-application-with-stm32cubemx-stmicroelectronics.pdf
& Software

Application Notes https://www.st.com/resource/en/application_note/an5426-migrating-
for related Tools graphics-middleware-projects-from-stm32cubemx-540-to-stm32cubemx-

& Software	550-stmicroelectronics.pdf
Application Notes for related Tools	https://www.st.com/resource/en/application_note/an5464-position-control-of-a-three-phase-permanent-magnet-motor-using-xcubemcsdk-or-xcubemcsdkful-stmicroelectronics.pdf
& Software	
Application Notes for related Tools	https://www.st.com/resource/en/application_note/an5564-getting-started-with-projects-based-on-dual-core-stm32wl-microcontrollers-in-stm32cubeide-stmicroelectronics.pdf
& Software	
Application Notes for related Tools	https://www.st.com/resource/en/application_note/an5698-adapting-the-xcubestl-functional-safety-package-for-stm32-iec-61508-compliant-to-other-safety-standards-stmicroelectronics.pdf
& Software	
Application Notes for related Tools	https://www.st.com/resource/en/application_note/an5731-stm32cubemx-and-stm32cubeide-threadsafe-solution-stmicroelectronics.pdf
& Software	
Application Notes for related Tools	https://www.st.com/resource/en/application_note/an5785-boost-voltage-mode-on-bg474edpow1-discovery-kit-stmicroelectronics.pdf
& Software	
Application Notes for related Tools	https://www.st.com/resource/en/application_note/an5788-stm32-digital-power-pid-and-iir-filters-for-smmps-control-design-and-comparison-on-bg414edpow1-discovery-kit-stmicroelectronics.pdf
& Software	
Application Notes for related Tools	https://www.st.com/resource/en/application_note/an4502-stm32-smbus-pmbus-expansion-package-for-stm32cube-stmicroelectronics.pdf
& Software	
Application Notes for related Tools	https://www.st.com/resource/en/application_note/an5952-how-to-use-cmake-in-stm32cubeide-stmicroelectronics.pdf
& Software	
Application Notes for related Tools	https://www.st.com/resource/en/application_note/an4635-how-to-optimize-lpuart-power-consumption-on-stm32-mcus-stmicroelectronics.pdf
& Software	
Application Notes for related Tools	https://www.st.com/resource/en/application_note/an5054-how-to-perform-secure-programming-using-stm32cubeprogrammer-stmicroelectronics.pdf
& Software	
Application Notes for related Tools	https://www.st.com/resource/en/application_note/an5496-guidelines-for-the-buck-voltage-mode-on-the-bg474edpow1-discovery-kit-stmicroelectronics.pdf
& Software	

Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5497-introduction-to-the-buck-current-mode-with-the-bg474edpow1-discovery-kit-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an6179-how-to-integrate-the-stl-firmware-into-a-time-critical-user-application-stmicroelectronics.pdf
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